

BAHARLY, Turkmenistan

WMO: 387740

Lat: 38.433N

Lon: 57.417E

Elev: 522

StdP: 14.42

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	23.7	11.8	10.2	27.1	17.6	13.6	32.8	18.4	51.4	13.9	51.2	3.6	110	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	105.3	71.5	102.9	70.7	100.7	70.3	76.9	97.7	75.1	96.1	73.5	94.4	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	113.1	91.2	67.7	103.8	90.5	65.7	96.6	88.6	40.7	98.1	38.9	95.9	37.4	94.5	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	11.7	9.6	DB	14.3	110.1	7.8	2.9	8.7	112.1	4.2	113.8	-0.2	115.5	-5.8	117.6	(4)
(5)			WB	13.7	80.4	8.1	4.3	7.9	83.5	3.2	86.0	-1.4	88.5	-7.2	91.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.6	39.6	42.5	53.1	64.3	76.8	86.1	90.0	88.0	77.4	64.0	50.3	41.5	(6)
(7)		DBStd	19.68	8.49	9.58	8.80	8.85	6.51	5.37	4.50	4.97	6.35	8.38	8.92	8.53	(7)
(8)		HDD50	1041	338	238	64	6	0	0	0	0	0	7	99	289	(8)
(9)		HDD65	3221	787	630	384	118	5	0	0	0	3	118	447	729	(9)
(10)		CDD50	6374	16	29	160	434	832	1083	1241	1178	823	442	110	26	(10)
(11)		CDD65	3077	0	1	15	96	371	633	776	713	376	88	7	1	(11)
(12)		CDH74	41341	0	7	195	900	3918	9041	11987	10459	4011	760	56	7	(12)
(13)		CDH80	23938	0	0	62	327	1778	5373	7753	6519	1873	239	14	0	(13)
(14)	Wind	WSAvg	3.5	3.2	3.5	3.8	3.6	3.8	4.0	3.7	3.4	3.2	3.1	2.9	3.2	(14)
(15)	Precipitation	PrecAvg	8.2	0.8	1.0	1.4	1.2	0.9	0.2	0.2	0.2	0.6	0.9	0.7		(15)
(16)		PrecMax	12.8	1.7	2.8	3.3	3.2	2.7	0.7	1.1	0.7	1.6	3.0	2.3	2.0	(16)
(17)		PrecMin	4.5	0.2	0.2	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	(17)
(18)		PrecStd	2.0	0.4	0.6	0.7	0.9	0.7	0.2	0.3	0.2	0.4	0.6	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	73.8	87.5	95.0	100.0	106.1	109.8	108.0	100.5	93.8	81.2	74.0	(19)
MCWB			50.7	53.6	61.2	66.3	68.7	70.8	74.3	71.8	68.6	63.8	60.7	54.5	(20)	
2%		DB	63.3	66.9	79.2	88.1	95.7	103.0	105.9	104.6	96.2	85.9	72.9	66.7	(21)	
		MCWB	49.4	51.0	58.3	63.8	67.5	70.1	72.5	70.6	67.6	63.1	56.8	51.1	(22)	
5%		DB	56.9	61.1	73.1	83.3	92.4	100.6	103.4	101.9	93.0	81.6	67.7	60.0	(23)	
		MCWB	47.1	49.2	56.4	62.2	66.4	69.4	71.5	70.5	66.7	61.6	54.7	48.5	(24)	
10%		DB	51.3	55.8	67.4	78.9	89.1	98.0	101.0	99.4	89.7	77.9	63.5	53.3	(25)	
		MCWB	44.8	47.3	53.9	60.5	65.6	69.0	71.0	70.2	65.9	59.9	53.2	46.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	55.5	64.0	68.3	73.3	77.1	80.5	78.6	73.4	68.4	61.5	55.7	(27)
MCD B			63.9	69.2	80.9	87.4	91.5	98.9	100.6	98.6	92.3	84.3	76.5	70.0	(28)	
2%		WB	50.3	52.5	59.6	65.9	70.2	74.0	77.3	76.5	71.2	65.5	58.7	52.0	(29)	
		MCD B	61.3	64.1	74.9	84.4	89.6	96.3	98.4	96.9	89.9	81.6	69.5	64.0	(30)	
5%		WB	47.7	50.0	57.1	63.7	68.6	72.4	75.5	74.7	69.5	63.3	56.5	49.1	(31)	
		MCD B	56.8	59.2	70.5	80.3	87.5	94.8	96.7	95.3	87.9	78.3	65.3	57.3	(32)	
10%		WB	45.1	47.7	54.8	61.7	67.2	70.9	73.9	72.8	67.8	61.3	54.2	46.9	(33)	
		MCD B	51.2	55.1	66.4	76.5	85.5	93.3	94.9	93.9	86.0	75.4	62.5	53.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.4	15.5	17.8	18.5	19.3	20.0	19.8	21.3	21.6	20.4	16.2	13.1	(35)
MCDBR			22.6	25.1	26.8	25.2	23.5	23.4	23.0	23.7	24.8	24.9	22.7	24.0	(36)	
MCWBR			12.4	12.8	12.1	9.9	9.3	9.3	8.6	9.8	10.7	10.7	11.0	12.4	(37)	
5% WB		MCDBR	20.2	22.3	23.5	22.2	20.8	20.1	18.9	20.3	20.8	21.6	19.0	19.7	(38)	
		MCWBR	12.1	12.3	11.7	10.1	9.5	10.0	8.7	10.4	10.4	10.4	9.9	11.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.413	0.467	0.548	0.614	0.610	0.602	0.618	0.560	0.526	0.535	0.464	0.418	(40)	
taud		2.019	1.883	1.731	1.653	1.692	1.714	1.688	1.787	1.836	1.802	1.937	2.007	(41)		
Ebn at Noon		227	230	224	220	225	226	221	230	228	206	206	213	(42)		
Edh at Noon		40	53	68	78	76	74	76	67	60	56	43	38	(43)		
(44)	All-Sky Solar Radiation	RadAvg	679	938	1316	1720	2138	2476	2414	2242	1837	1265	835	609	(44)	
(45)		RadStd	68	110	100	166	106	73	82	96	57	78	63	76	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.86	-1.21	-1.17	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	N/A	+148

Nomenclature: See separate page